

Retail Sales Analysis Case Study

Project Overview

This case study presents a **Retail Sales Analysis** project developed using **SQL, Microsoft Excel, and Power BI**. The project analyzes **2,000 retail sales transactions** to evaluate sales performance, product profitability, customer purchasing behavior, and regional sales trends.

The analysis includes SQL-based business queries, customer segmentation, retention analysis, KPI reporting, and interactive dashboards to identify sales patterns, monitor business performance, and support data-driven decisions related to products, customers, and revenue growth.

Analysis Objective

The objective of this project was to analyze banking data to:

- Analyze overall sales and profit performance.
 - Identify top-performing products and product categories.
 - Evaluate customer purchasing behavior through customer segmentation and retention analysis.
 - Monitor sales trends across cities and time periods.
 - Develop KPI reports and interactive dashboards for business reporting.
 - Generate actionable insights to support sales optimization and business growth.
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Business Problem

The retail company generated large volumes of sales data but lacked meaningful reporting to monitor sales performance, customer purchasing behavior, and product profitability. Decision-makers needed a centralized solution to identify high-performing products, understand customer trends, evaluate repeat purchases, and track business KPIs for better operational and strategic decisions..

Dataset Description

- **Total Orders:** 2,000
- **Total Customers:** 738
- **Product Categories:** 6

- **Sales Channels:** Multiple (Online, Store, etc.)
- **Cities Covered:** 7

Key Attributes

- Order_ID
- Order_Date
- Customer_ID
- Product_Name
- Category
- Quantity
- Unit_Price
- Revenue
- Cost
- Profit
- Payment_Method
- Sales_Channel
- Customer_Rating
- Customer_Type
- Discount_Percentage
- City

Data Dictionary

Order_ID	Unique identifier assigned to each retail order.
Order_Date	Date on which the customer placed the order.
Customer_ID	Unique identifier assigned to each customer.
Product_Name	Name of the product purchased by the customer.
Category	Product category used to analyze sales and profitability.
Quantity	Number of units purchased in each order.
Revenue	Total sales revenue generated from the order.
Profit	Profit earned after deducting the product cost from the revenue.
Sales_Channel	Channel through which the product was sold (Online, Store, etc.).
Customer_Type	Classification of customers based on their purchasing behavior.

Tools & Technologies Used

- **Database:** MySQL – Data storage and business query analysis.
 - **SQL:** Joins, Aggregate Functions, CASE Statements, CTEs, Window Functions, Customer Segmentation, Retention Analysis.
 - **Data Handling:** CSV Files – Data import, validation, and preprocessing.
 - **Microsoft Excel:** Data Cleaning, Pivot Tables, Pivot Charts, KPI Reporting, Business Reporting.
 - **Power BI:** Data Modeling, DAX Measures, KPI Cards, Interactive Dashboards, Data Visualization, Business Intelligence Reporting.
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Data Preparation & Preprocessing

The retail sales dataset was prepared before analysis to ensure accurate, consistent, and reliable reporting across SQL, Excel, and Power BI.

- Removed duplicate records to maintain data integrity.
 - Identified and handled missing or invalid values where applicable.
 - Corrected data types for dates, numeric fields, and categorical columns.
 - Standardized column names and data formatting for consistency.
 - Validated sales calculations, including **Revenue**, **Cost**, and **Profit**.
 - Verified customer, product, and order information before performing business analysis.
 - Imported the cleaned dataset into **MySQL**, **Microsoft Excel**, and **Power BI** for reporting and dashboard development.
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Analysis Performed

The retail sales dataset was analyzed using **SQL**, **Microsoft Excel**, and **Power BI** to evaluate sales performance, customer behavior, and product profitability, and to generate actionable business insights.

1. Sales Performance Analysis

- Analyzed total revenue, profit, total orders, and average order value.
- Identified monthly sales trends and overall business performance.

2. Product Performance Analysis

- Ranked products based on revenue using SQL Window Functions.
- Compared revenue and profit across different product categories.
- Identified top-performing and low-performing products.

3. Customer Analysis

- Segmented customers into High, Medium, and Low Value groups using SQL.
- Performed customer retention analysis to identify repeat customers.
- Analyzed customer distribution and purchasing behavior.

4. Regional Sales Analysis

- Compared revenue across different cities.
- Identified cities contributing the highest sales.

5. Dashboard & KPI Analysis

- Developed KPI reports and interactive dashboards using Excel and Power BI.
- Monitored key business metrics including Revenue, Profit, Orders, Customers, and Average Order Value.
- Visualized sales trends, category performance, city-wise sales, and customer insights to support business decision-making.

Key Observations & Business Relevance

Key Observations

- Fashion category generated the highest revenue, followed closely by Home and Beauty products.
- Revenue and profit varied across product categories, highlighting opportunities to optimize the product mix.
- Pune recorded the highest overall sales revenue among all cities.
- Monthly sales analysis showed fluctuations in revenue, with March recording the highest sales.
- Customer segmentation identified High, Medium, and Low Value customers based on total spending.
- Retention analysis showed that a significant portion of customers made repeat purchases, indicating customer loyalty.
- KPI dashboards provided a consolidated view of sales performance, profitability, customer metrics, and regional trends.

Business Relevance

The analysis provides actionable insights that support retail business decisions by:

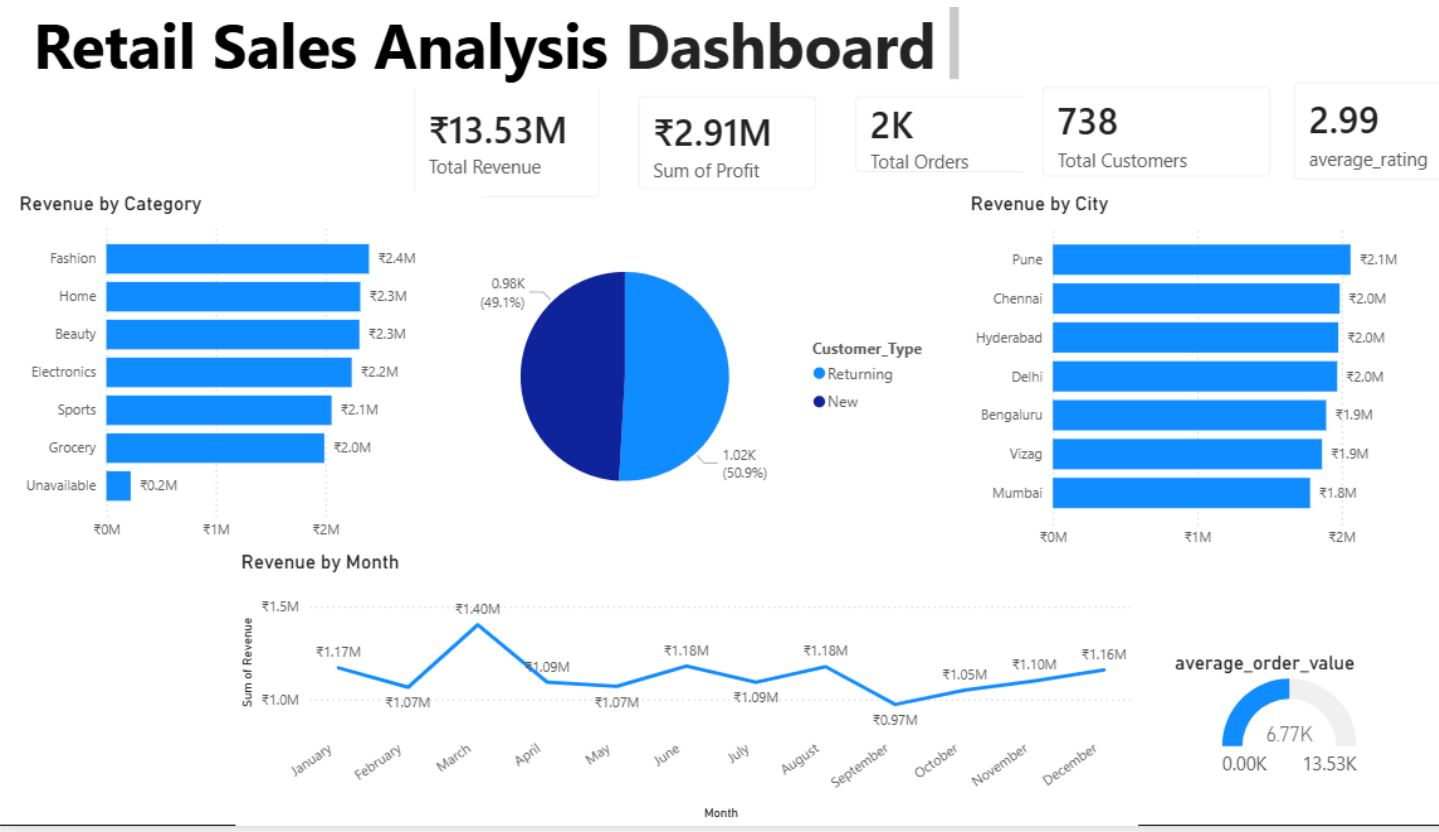
- Identifying high-performing products and categories for better inventory planning.
- Understanding customer purchasing behavior to improve customer retention and marketing strategies.
- Monitoring city-wise sales performance to optimize regional business operations.
- Tracking monthly sales trends to support sales forecasting and promotional planning.

- Enabling management to monitor key business KPIs through interactive dashboards for faster and data-driven decision-making.

Retail Sales Analysis Project | SQL, Microsoft Excel, Power BI

- Performed end-to-end analysis on **2,000 retail sales transactions** to evaluate sales performance, product profitability, and customer purchasing behavior.
- Developed **SQL business solutions** using **Joins, Aggregate Functions, CASE Statements, CTEs, Window Functions, Customer Segmentation, and Retention Analysis** to generate business insights.
- Built **Excel KPI reports**, Pivot Tables, and Pivot Charts to analyze revenue, profit, sales trends, and regional performance.
- Designed an interactive **Power BI dashboard** using **Data Modeling, DAX Measures, KPI Cards, and visualizations** to monitor key retail business metrics.
- Analyzed product categories, city-wise sales, monthly revenue trends, and customer behavior to support business decision-making.
- Generated actionable business recommendations to improve sales performance, customer retention, product strategy, and overall retail operations.

Power BI:



Sheets:

Payment_Metho	COUNTA of Customer_ID
UPI	415
Debit Card	412
Cash	396
Credit Card	392
Wallet	385

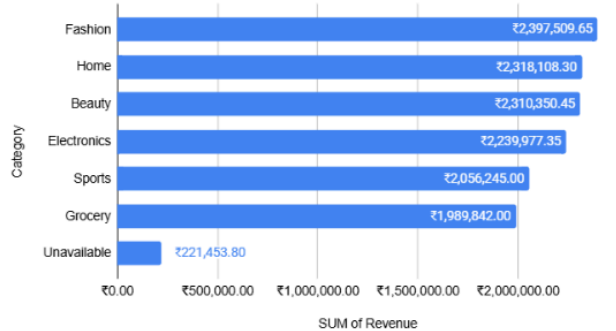
City	SUM of Revenue
Pune	₹2,061,713.75
Chennai	₹1,986,443.40
Hyderabad	₹1,976,945.80
Delhi	₹1,969,525.00
Bengaluru	₹1,892,941.25
Vizag	₹1,864,201.25
Mumbai	₹1,781,716.10

Category	SUM of Revenue
Fashion	₹2,397,509.65
Home	₹2,318,108.30
Beauty	₹2,310,350.45
Electronics	₹2,239,977.35
Sports	₹2,056,245.00
Grocery	₹1,989,842.00
Unavailable	₹221,453.80

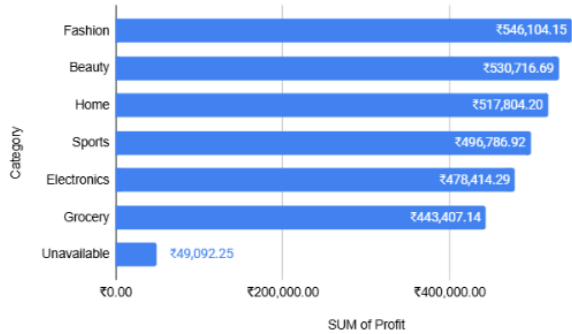
Category	SUM of Profit
Fashion	₹546,104.15
Beauty	₹530,716.69
Home	₹517,804.20
Sports	₹496,786.92
Electronics	₹478,414.29
Grocery	₹443,407.14
Unavailable	₹49,092.25

Total Revenue	₹13,533,486.55
Total Profit	₹3,062,325.64
Total Orders	2000
Total Customers	2001
Average Rating	3
Average Order V	₹6,766.74

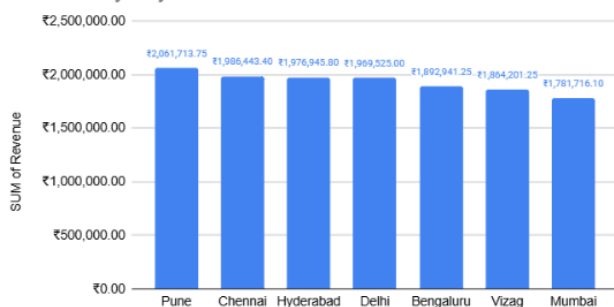
Revenue by Category



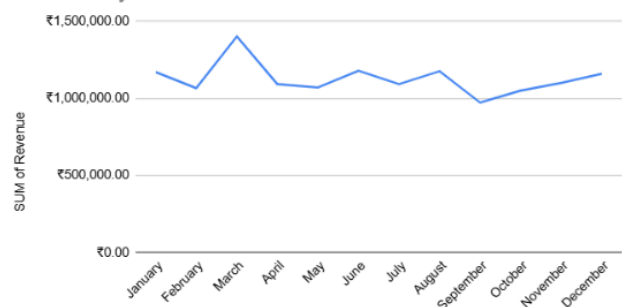
Profit by Category



Revenue by City



Revenue by Month



Conclusion

This **Retail Sales Analysis** project provided valuable insights into sales performance, customer purchasing behavior, product profitability, and regional sales trends using **SQL, Microsoft Excel, and Power BI**. Through business-focused SQL analysis, KPI reporting, customer segmentation, retention analysis, and interactive dashboards, the project transformed retail sales data into meaningful business insights.

The analysis helped identify top-performing products, monitor key business metrics, evaluate customer behavior, and support data-driven decisions for sales optimization and business growth. Overall, this project strengthened my practical skills in **data analysis, business intelligence, dashboard development, and retail analytics**, while demonstrating my ability to convert raw business data into actionable insights for decision-makers.